



Computer Networks 05 | Problem Solving on Subnetting (Part-1) | CS & IT | GATE Crash Course

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Summary

AI Notes

AI Slides

Text Notes

Screenshots

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Q.9 Consider default subnet mask for a network is 255.255.255.0.
How many number of hosts per subnet possible if 'm' bits are borrowed from Host ID (HID)

A $2^{HID-m}-2$ ✓ *class C*

B 2^{HID} *$\frac{HID}{24} \quad \frac{HID}{8}$*

C 2^{HID-m} *$\frac{m}{SID} \quad \frac{HID-m}{HID}$*

D 2^m

No. of Host/subnet = $2^{HID-m} - 2$

GATE WALLAH

▶ 4:54

Q.10

A university has Subnets, with 100 hosts in each subnet. If it uses class B then the subnet mask in Dotted Decimal Notation is _____.

class-B

$$\frac{NID}{16} \quad \frac{HID}{16}$$

$$\frac{NID}{16} \quad \frac{SID}{9} \quad \frac{HID}{7}$$

No. of 1's in the subnet Mask = $NID + SID = 16 + 9$
 No. of 0's in the " " = $HID = 7$

||||| · ||||| · ||||| · 1000

255.255.255.128

▶ 9:10

Q.11

A university has 150 Subnets. Use Class B address and then the subnet mask in Dotted Decimal notation is _____.

class-B

$$\frac{NID}{16} \quad \frac{HID}{16}$$

150 subnet

$$\frac{16}{NID} \quad \frac{8}{SID} \quad \frac{8}{HID}$$

No. of 1's in the S.M = $NID + SID + 8 = 24$

" " 0's " " " " = HID

||||| · ||||| · ||||| · 0

255.255.255.0

▶ 12:23

Q.1

IP Address = 200.200.200.126

Subnet Mask = 255.255.255.192

Then Find SID and HID

$$\begin{aligned} \text{HID} &= \text{IPAdd} - \text{SID} \\ &= 126 - 64 \\ \text{HID} &= 62 \end{aligned}$$

IP Address

ANDing

Network Mask

NID

IP Address = 200.200.200.0 1111110

ANDing AND

Subnet Mask = 255.255.255.1 100000

SID = 200.200.200.0 100000

SID = 200.200.200.64

HID = 200.200.200.62

▶ 18:12

Subnet Mask = 255.255.255.224

$$= \begin{array}{cccc} 11111111 & 11111111 & 11111111 & 11100000 \\ & & & \text{HID} \end{array}$$
~~No of 1's = 27~~~~NID + SID = 27~~~~24 + SID = 27~~~~SID = 3 bit~~

No. of 0's = 5

HID = 5 bit

✓

▶ 31:05

Q.5

If subnet Mask = 255.255.255.224 then Find

- (i) No. of IP Address/subnet possible 2^5
 (ii) No. of Host/subnet possible $2^5 - 2$
 (iii) No. of subnet in class A _____
 (iv) No. of subnet in class B _____
 (v) No. of subnet in class C _____

class-A

$$\begin{aligned} \text{NID} + \text{SID} &= 24 \\ 8 + \text{SID} &= 24 \\ \text{SID} &= 16 \text{ bit} \end{aligned}$$

No. of subnet in class A = 2^{16}

class-C

$$\begin{aligned} \text{NID} + \text{SID} &= 24 \\ 24 + \text{SID} &= 24 \\ \text{SID} &= 0 \text{ bit} \end{aligned}$$

No. of subnet in class C = 2^0

class-B

$$\begin{aligned} \text{NID} + \text{SID} &= 24 \\ 16 + \text{SID} &= 24 \\ \text{SID} &= 8 \text{ bit} \end{aligned}$$

No. of subnet in class B = 2^8

11111111.11111111.11111111.11100000

10.0F 1's = 24

NID + SID = 24

No. of 0's = 5

HID = 5 bit

No. of IP Addr/subnet = $2^5 = 32$ No. of Host/subnet = $2^5 - 2 = 30$

GATEWAY

▶ 36:27

Q.6

If subnet Mask = 255.255.255.240 then Find

- (i) No. of IP Address/subnet possible = 2^4
 (ii) No. of Host/subnet possible = $2^4 - 2$
 (iii) No. of subnet in class A = 2^{20}
 (iv) No. of subnet in class B = 2^{12}
 (v) No. of subnet in class C = 2^4

Sm: 11111111.11111111.11111111.11110000
 class A
 class B
 class C
 HID

▶ 43:11

Q.9 If subnet mask is 255.255.224.0 then number of subnet is

msg

☒ A 2^{11}

☐ B 2^5

☐ C 2^{15}

☒ D 2^3

AD

class A
class B

||||| · ||||| · || 0000 · 00000000

PW⁵¹

GATEWAY

▶ 51:07

Q.12 If subnet mask is 255.255.255.248, then which of the following can be Direct broadcast address

msg

☒ A 200.32.64.135 [1 0000 111]

☒ B 200.32.64.240 [1111 0000]

☒ C 200.32.64.207 [1100 1111]

☒ D 200.32.64.231 [11100 111]

sm: 255.255.255.11111000
HID

HID=3bit

ACD

PW⁰²

GATEWAY

▶ 1:02:16

Q.17 IP address in a block = 200.200.200.122 and the subnet Mask = 255.255.255.240 then Find

(i) Subnet id = 200.200.200.112

(ii) Subnet number = 3rd subnet

Sm: $\begin{array}{c} \text{|||||} \cdot \text{|||||} \cdot \text{|||||} \cdot \text{||||} \text{0000} \\ \text{NID} \quad \text{SID} \quad \text{HID} \end{array}$

$200.200.200. \begin{array}{c} \text{0111} \\ \text{SID} \end{array} \text{----} \text{HID}$

$200.200.200. \begin{array}{c} \text{0111} \\ \text{SID} \end{array} \text{0000} \rightarrow 200.200.200.112 \text{ SID}$

$\text{SID} \text{ dec } \rightarrow 7 \rightarrow 8^{\text{th}} \text{ subnet}$

▶ 1:29:22

Q.19 IP address in a block = 157.157.52.80 and the subnet Mask = 255.255.192.0 then Find

(i) Subnet-id ____.

(ii) First host ____.

(iii) Last host ____.

(iv) Direct broadcast address ____.

Sm: $\begin{array}{c} \text{|||||} \cdot \text{|||||} \cdot \text{||} \text{000000} \cdot \text{00000000} \\ \text{NID} \quad \text{SID} \quad \text{HID} \end{array}$

$157.157. \begin{array}{c} \text{00} \\ \text{SID} \end{array} \text{-----} \text{HID}$

$157.157. \begin{array}{c} \text{00} \\ \text{SID} \end{array} \text{000000} \cdot \text{00000000} \rightarrow 157.157.0.0$

$157.157. \begin{array}{c} \text{00} \\ \text{SID} \end{array} \text{000000} \cdot \text{00000001} \rightarrow 157.157.0.1$

$157.157. \begin{array}{c} \text{00} \\ \text{SID} \end{array} \text{|||||} \cdot \text{|||||} \cdot \text{0} \rightarrow 157.157.63.254$

$157.157. \begin{array}{c} \text{00} \\ \text{SID} \end{array} \text{|||||} \cdot \text{|||||} \cdot \text{1} \rightarrow 157.157.63.255$

▶ 1:44:24